Shill Bidding in MEV-Boost Auctions

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Abstract

Ethereum's Proposer-Builder Separation allows validators (proposers) to outsource block-building to specialized builders via MEV-Boost auctions. However, exclusive order flow deals have reduced builder competition, causing significant revenue losses for proposers.

This paper introduces a novel strategy called shill bidding for proposers to counteract such revenue losses, in which a proposer impersonates builders and submits fake bids to artificially inflate final auction prices. Shill bidding is technically simple and virtually risk-free, allowing proposers to increase their revenue without the need for profitable order flows or sophisticated block-building techniques. To analyze the effects of shill bidding, we propose a novel auction model with probabilistic termination. Our theoretical analysis and numerical results indicate that shill bidding can consistently bring more revenue, with the magnitude of the gain depending on the relative strength of the top builders. In addition, it may amplify proposers' incentives to engage in timing games. Empirical results suggest that shill bidding could potentially yield thousands of ETH in additional monthly revenue, highlighting strong incentives for its adoption.

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1 Introduction

A central challenge in modern blockchain systems is the extraction of economic value from transaction ordering, a phenomenon known as Maximal Extractable Value (MEV) [5]. MEV is defined as the profit that can be obtained by including, excluding, and reordering transactions within a block. Validators, who have the final say on what goes into a block in the MEV supply network, naturally have the advantage of capturing MEV. Maximizing MEV typically requires sophisticated technical knowledge and specialized infrastructure integrated into normal validator operations. Consequently, validators with superior resources and access to advanced MEV extraction techniques often consistently outperform solo validators, creating a centralizing force.

To solve the centralization risk posed by MEV on validators, Ethereum has introduced an architecture for block production called Proposer-Builder Separation (PBS) [6], separating the role of building blocks from proposing them. Under PBS, the task of block construction is outsourced to a market of builders. Specialized block builders handle the complexity of constructing blocks that maximize MEV. Each slot, the proposer (i.e., a randomly chosen validator) selects the most profitable one among multiple block candidates submitted by competing builders and proposes it as the new head of the blockchain. As a result, validators benefit from MEV without directly engaging in the MEV extraction process themselves.

Currently, the idea of PBS is implemented via an out-of-protocol solution referred to as *MEV-Boost*. The block production process has also evolved into an intricate interplay between specialized actors: *Searchers* operate at the transaction level and look for MEV opportunities (e.g., arbitrages [32, 12], liquidations [21], sandwiches [36]) to bundle transactions together to earn MEV. They send bundles of transactions alongside payments to express their order preference to one or multiple block builders. *Builders*, operating at the block level, pack their received searcher bundles and normal user transactions to construct maximally profitable blocks. They compete by submitting blocks alongside bids, while the *proposer* of the slot chooses the highest-bid block to include on-chain. In a fully competitive scenario where builders are evenly matched, the proposer is expected to capture most of the MEV via MEV-Boost auctions.

However, the phenomenon of integration has led to significant inequality in block-building capability among builders. A builder's capability to produce competitive blocks largely depends on the profitability of the transactions and MEV bundles (collectively referred to as order flows) they can access. In practice, certain pivotal order flow providers tend to exclusively channel their profitable order flows to specific builders, a strategy known as integration. When it comes to inequality of block-building capacities, two integration patterns are particularly worrying: first, searchers who seek arbitrage opportunities between centralized exchanges (CEXs) and decentralized exchanges (DEXs) also operate their own builders and exclusively send order flows internally [10]; second, leading independent providers such as Banana Gun, jaredfromsubway.eth, and Maestro, exclusively send order flows to specific external builders – Banana Gun to Titan, and both jaredfromsubway.eth and Maestro to Beaver [34]. These exclusive arrangements create disparities in builders' competitive capabilities. Due to the integration, as of the time of writing, the top two builders, Titan and Beaver, consistently control over 70-90% of the market [27]. Although both with exclusive order flow providers, their abilities within the same slot may vary greatly.

As a result, proposers face significant losses [34], especially when an order flow is overwhelmingly profitable and only accessible to a builder. Other builders, lacking access to the high-value order flows, are unable to compete and submit competitive bids. This allows the winning builder to bid only marginally above the second-highest offer, capturing most of the MEV for themselves and their order flow providers. The proposer, who ideally should benefit from intense bidding competition, instead receives a fraction of the potential surplus. In the long run, such a loss incentivizes proposers to return to the pre-PBS world and build blocks themselves, violating the goal of PBS.

1.1 Our Contributions

In this paper, we examine what proposers can do to increase revenue in MEV-Boost auctions to counteract such losses. We summarize our main contributions in the following.

- We identify a strategy called *shill bidding*, wherein a proposer in MEV-Boost auctions can pretend to be a builder and submit fake bids to artificially inflate the final auction price. We show that this strategy is easy to execute, virtually risk-free, and difficult to detect, allowing proposers to increase their revenue without the need for profitable order flows or sophisticated block-building techniques. We discuss shill bidding in the MEV-Boost auction in Section 2.

- To analyze the effectiveness of shill bidding, we propose a novel auction model which we term *auction with probabilistic termination*. We consider the strategic interaction among two builders[1] and a single proposer, and characterize the Bayesian subgame-perfect equilibrium strategies of the builders. We further show how shill bidding can mislead real builders into perceiving greater competition than actually exists, thereby altering their bidding behavior. We derive the integral form of the proposer's expected revenue and conduct numerical studies to evaluate the impact of shill bidding. Among several notable findings in Section 3.5, we highlight a particular concern: shill bidding may exacerbate the timing game activity.
- Our empirical study leveraging historical data shows that shill bidding can significantly increase proposer revenue, e.g., over 7,000 ETH in a single month. We also develop heuristic methods to empirically detect signs of shill bidding. While we do not observe conclusive evidence of its current adoption, the magnitude of potential gains suggests that proposers may be incentivized to adopt such behavior in the future.

1.2 Related work

Proposer-Builder Separation. Heimbach et al. [11] studied the extent to which PBS achieves its goal of enhancing decentralization. They found that PBS effectively levels the playing field by enabling all validators, regardless of size, access to competitive blocks, thus preventing solo validators from being outcompeted by large ones. Several works [35, 11, 29] have also analyzed the PBS's impact on censorship resistance.

Strategic Behavior in MEV-Boost Auctions. [25, 33] investigated diverse bidding behaviors among builders leveraging empirical bid data. The role of latency has been emphasized in shaping auction outcomes [19]: builders with lower latency benefit from information advantages, enabling them to respond faster and submit bids later. Other works [10, 18, 34, 30] have shown that access to profitable order flow is a key determinant of builder success and contributes to the centralization of the builder market. On the proposer side, Schwarz-Schilling et al. [24] proposed timing games, showing that proposers have the incentive to delay block publication for as long as possible to capture higher MEV. Neuder et al. [15] stated that the design of bid cancellations (details in Section 2.1) incentivizes proposers to ask for the highest bid repeatedly and only sign the overall highest one.

Shill Bidding. The practice of shill bidding or shilling refers to the participation of sellers in their own auctions to inflate prices [2]. It has been frequently observed [17, 3] in traditional auctions, especially in online platforms such as eBay, where sellers can easily conceal their identity. Recent theoretical work by Komo, Kominers, and Roughgarden characterized auction formats that are shill-proof, i.e., those in which profit-maximizing sellers have no incentive to place shill bids. In particular, they have shown that the Dutch auction (with a suitable reserve) is the unique (revenue-)optimal and strongly shill-proof auction, while the English auction with an optimally chosen reserve price is weakly shill-proof, optimal, and strategy-proof. In blockchain settings, the literature on transaction fee mechanism design (e.g., [23, 1, 4, 8, 9]) has explored analogous concerns, where miners may boost their revenue through creating (and including) fake transactions, which can be viewed as a form of shill bidding. To the best of our knowledge, our work is the first to study shill bidding in the context of MEV-Boost auctions.

[1] This is motivated by recent empirical evidence [27] showing that Titan and Beaver have consistently dominated the MEV-Boost auction.

2 Shill Bidding

This section first provides an overview of MEV-Boost auctions, emphasizing their distinct features, and then describes the shill bidding strategy in detail within this auction framework.

2.1 MEV-Boost auctions

MEV-Boost auctions enable the validator of each slot (i.e., the proposer) to access blocks from a builder market through trusted intermediaries known as *relays*, and select the most profitable block to propose.

In each slot, builders compete for the right to land their block on-chain, which is auctioned off by the proposer. Each builder continuously tries to construct more profitable blocks and submits a bid – indicating their proposed payment to the proposer – along with the corresponding block to the relay. The relay coordinates between builders and proposers. It verifies that the received block is valid and that the builder has sufficient balance to pay their promised bid. Only the submission that passes the verification will be considered eligible. From the relay's perspective, each public key is an entity. A builder may use multiple public keys to participate. The relay maintains the latest eligible bid and corresponding block for each public key. Any new submission will overwrite a previous one by the same public key, even if it is less profitable. Introducing the relay ensures that proposers cannot access the block contents before committing to proposing the block, thereby preventing theft or censorship of MEV opportunities. At the end of the auction, the proposer queries the relay for the highest bid until that time via a `getHeader` call. It then signs the block header as a commitment to proposing that block and returns it to the relay by calling `getPayload`. Upon receiving the signed block header, the relay publishes the full block contents to the proposer and the peer-to-peer network, finalizing the auction for the slot.

The bidding process in the MEV-Boost auction is partially observable: anyone, including both the proposer and builders, can repeatedly query the relay during the auction via `getHeader` to monitor the current highest bid. Builders can use this feedback to adjust their bids in real time. Thus, the mechanism resembles an open English auction, although with several distinguishing features:

(1) **Time-bounded finalization:** The auction is constrained by block time, which is 12 seconds in Ethereum. Specifically, the proposer determines the exact end time of the auction by sending the signed block header as a signal. The block auction for a slot begins in the previous slot. An honest proposer is expected to conclude the auction at the beginning of the slot (i.e., $t = 0$ s of the slot), leaving sufficient time for the following stage of attestation, where a selected committee of attesters from the validator set votes for the head of the canonical blockchain.

(2) **Dynamic valuations:** Unlike traditional auctions where bidders' true values are typically static, in MEV-Boost auctions, a builder's true value equals the real-time MEV revenue of their block. As available order flow changes and new block candidates are constructed, the block value evolves dynamically during the auction, with a trend of increasing over time. As a result, proposers tend to strategically delay their block proposal as long as possible while still meeting the attestation deadline ($t = 4$ s of the slot). This delay gives builders extra time to enhance their blocks, leading to more MEV for the proposer. Currently, proposers, especially sophisticated ones like large staking pools, usually delay concluding the auction at $t = 2 - 3$ s of the slot [26].

(3) **Bid cancellation:** Builders can effectively withdraw their previous bids by submitting lower ones later in the auction period, which is very rare in traditional auctions. This

flexibility is designed to accommodate the uncertainty in the profitability of certain MEV opportunities. For instance, the profit from a CEX-DEX arbitrage may decrease as CEX prices update. In response, searchers may lower their payment to the builder or cancel their bundles, reducing the value of the builder's block. Allowing bid cancellation enables builders to adjust accordingly to such changes in real time.

2.2 Shill bidding in MEV-Boost auctions

Shill bidding refers to the behavior where the seller (or their accomplice) places fake bids on their own auction to artificially inflate the final price. This behavior manipulates the bidding process and misleads real bidders into believing there is more intense competition in the item than there actually is, causing them to bid more than they otherwise would.

▶ **Example 1.** Consider an online English auction. In an English auction, bidding starts at a low price, and participants publicly place progressively higher bids until no one is willing to bid further; the highest bidder wins the item at their final bid price. Suppose the seller lists an item with a starting price of \$200. A real bidder who values the item at \$300 places a bid of \$220. If no other bidders participate, the auction would naturally end at \$220. However, in an attempt to increase revenue, the seller uses another account to place a fake bid of \$260. As a response, the real bidder raises their bid to \$280. The shill bidder (i.e., the seller) then drops out, and the real bidder wins at an inflated price of \$280, paying more than they otherwise would have.

In MEV-Boost auctions, a simple strategy for proposers to increase revenue is shill bidding, namely, the proposer impersonates a builder to submit fake bids to encourage the real builders to go higher. Since the proposer can observe the current highest bid by repeatedly querying the relay, they can use a separate public key (different from their public key as the proposer) to submit a slightly higher fake bid to the relay. By incrementally pushing the price upward, proposers artificially inflate the auction outcome and increase their own revenue.

Technically, this shill bidding strategy is easy to implement. The proposer needs only to submit a valid block and bid. Constructing a high-value block is not necessary. Instead, the proposer can simply collect some public transactions/bundles and arrange them arbitrarily as a block – even if the block's true value is far below the stated bid. Overbidding does not appear suspicious in MEV-Boost auctions; in fact, builders often practice the so-called "block subsidization," namely, deliberately placing bids that exceed the block's extracted value to maintain or grow their market share and attract future private order flows [18, 34]. The only technical requirement for the proposer is to put sufficient funds in the shill bidder's account to make shill bids eligible, thereby passing the relay's validity checks. Thus, there is almost no barrier – both large and solo validators can easily adopt shill bidding.

Practically, shill-bidding is riskless in MEV-Boost auctions. In traditional auctions, shill bidders carry the significant risk of accidentally outbidding all real bidders and winning their own sale. For instance, consider the earlier Example 1 but assume that the shill bidder places a bid of \$320, the real bidder will drop out as the price has exceeded their maximum willingness to pay (i.e., \$300). In this case, the shill bidder (i.e., the seller) becomes the highest bidder and would be forced to "buy" their own item – an outcome that defeats the purpose of making a real sale and may also incur listing or final value fees from the auction platform. By contrast, in MEV-Boost auctions, the proposer (effectively the seller) has much more control: The proposer chooses which block header to sign, so they can ensure that the winning bid always comes from a real builder – even if a fake bid was the highest one.

Specifically, every time they query the relay, the proposer can locally record the real builders' highest bid so far and the corresponding block header, and eventually sign it if the relay's final response is a shill bid. Moreover, recall that the proposer also decides when to end the auction. A cautious proposer could even lower their bid just before finalizing the auction to lose intentionally. In this way, the proposer signs the final response from the relay as suggested, while still manipulating auction results.

Additionally, shill-bidding is difficult to detect in MEV-Boost auctions. In addition to lowering the bid at the last second, a proposer can switch multiple builder identities (public keys) during the adaptive bidding process to evade detection. As a result, some heuristic rules aimed at detecting shill bidding, such as identifying a new builder repeatedly bidding on a specific proposer's slots but rarely winning, may perform less effectively than expected.

3 Theoretical Model and Analysis

3.1 Model

We investigate the strategic interaction among block builders and a single proposer. In particular, we focus on a setting where only two builders, builder $\mathcal{B}$ and builder $\mathcal{T}$, are competing for their block to be selected. Each builder constantly constructs new blocks and submits a bid for each block. We assume the builder's bidding strategy is consistent across all the blocks they generate. Namely, there is a bidding function $b_{\mathcal{B}}(\cdot)$ and the builder $\mathcal{B}$ will bid $b_{\mathcal{B}}(v)$ for any block that is worth v for the builder $\mathcal{B}$. Similarly, there is a bidding function $b_{\mathcal{T}}(\cdot)$ and the builder $\mathcal{T}$ will bid $b_{\mathcal{T}}(v)$ for any block that is worth v for the builder $\mathcal{T}$. Thus, and naturally, the effective block for the builder $\mathcal{B}$ is the block with the highest valuation for $\mathcal{B}$, and we denote its valuation by $v_{\mathcal{B}}$. And the effective block for the builder $\mathcal{T}$ is the block with the highest valuation for $\mathcal{T}$, and we denote its valuation by $v_{\mathcal{T}}$.

Crucially, since each builder may generate multiple blocks, each of the builders $\mathcal{B}$ and $\mathcal{T}$ may submit multiple bids, and they may use different (and potentially newly created) public keys for each bid. Thus, in our setting, given a bid, a builder can only recognize if the bid is from themselves or not. Naturally, if they see a bid not from themselves, they will view this bid as the other builder's.

A novel feature in our model is a new auction format, which we term *auction with probabilistic termination*. This is especially motivated by the recent discrepancy in termination time in the MEV-Boost auction. We describe the auction as follows.

Each builder has a quasi-linear utility function – defined as their valuation minus their payment – and a valuation independently drawn from a distribution. Specifically, we assume the builder $\mathcal{B}$'s valuation $v_{\mathcal{B}} \sim \mathcal{U}[0, H_{\mathcal{B}}]$ and the builder $\mathcal{T}$'s valuation $v_{\mathcal{T}} \sim \mathcal{U}[0, H_{\mathcal{T}}]$. For simplicity and technical reasons, we assume that there is a $z \in [0, 1)$ such that $H_{\mathcal{T}} = \frac{1}{1+z}$ and $H_{\mathcal{B}} = \frac{1}{1-z}$. The parameter z is intended to quantify the relative power between the two major builders: the greater the value of z, the more pronounced the disparity in their capabilities of extracting MEV.

The auction proceeds in at most two rounds. At the outset, each builder simultaneously submits the bids, where the highest bids from $\mathcal{B}$ and $\mathcal{T}$ are denoted by $b^1_{\mathcal{B}}$ and $b^1_{\mathcal{T}}$, respectively. With probability $p \in (0, 1)$, the auction terminates immediately after the first round: the builder with the higher bid wins and pays their bid. With the remaining probability $1 - p$, the auction proceeds to a second round.

If the auction advances, all the bids from the first round are publicly revealed. In particular, the builder $\mathcal{T}$ can see $b^1_{\mathcal{B}}$ and the builder $\mathcal{B}$ can see $b^1_{\mathcal{T}}$. This provides a common informational update[2]. In the second round, after seeing the released information, both builders simultaneously submit updated bids, $b^2_{\mathcal{B}}$ and $b^2_{\mathcal{T}}$. The builder with the higher second-round bid wins the item and pays their bid.

▶ Remark 2 (Model explaination). First, the two-builder setting is motivated by recent empirical evidence [27] showing that Titan and Beaver have consistently dominated MEV-Boost auctions, often collectively winning 70-90% slots, primarily due to exclusive integrations with different major order flow providers. Besides, from the proposer's perspective, the revenue extracted through MEV-Boost auctions mainly hinges on the capability gap between the top two builders, making it reasonable to focus on their strategic interaction. Second, since it is the proposer who determines the end time, the auction duration is not deterministic. The historical data allows one to estimate the distribution of a validator's decision on the end time, motivating the use of probabilistic termination in our model.

3.2 Equilibrium analysis of the builders

How should the proposer predict the bidding behavior of the builders? Indeed, it might be too optimistic to assume that the builders bid truthfully according to their valuations, as such naive bidding strategies always yield zero utility to both builders. Instead, a more appropriate framework is provided by auction theory, specifically the concept of the Bayesian Nash equilibrium. Given our multi-round setting, we adopt the solution concept of the Bayesian subgame-perfect Nash equilibrium, which requires optimal behavior in every subgame, including after the first round when additional information is revealed.

We focus on *strictly increasing bidding strategies*, a common class of strategies in auction theory. Namely, we focus on the case where the builder $\mathcal{B}$ adopts a bidding strategy $b^1_{\mathcal{B}}(v_{\mathcal{B}})$ that is strictly increasing with respect to $v_{\mathcal{B}}$, and similarly, the builder $\mathcal{T}$ adopts a bidding strategy $b^1_{\mathcal{T}}(v_{\mathcal{T}})$ that is strictly increasing with respect to $v_{\mathcal{T}}$. Indeed, when the auction has a single round and two bidders with valuations uniformly distributed from two intervals, it is known that there is a unique Bayesian Nash equilibrium that is strictly increasing [13].

If the auction proceeds to a second round, which occurs with probability $1 - p$, then the bids from the first round are publicly revealed. This disclosure reveals the underlying valuations since the bidding strategy is strictly increasing. In particular, if the builder $\mathcal{B}$ sees the highest bid that is not from themselves is $\beta_{\mathcal{T}}$, then the builder $\mathcal{B}$ infers that $v_{\mathcal{T}} = (b^1_{\mathcal{T}})^{-1}(\beta_{\mathcal{T}})$. Similarly, if the builder $\mathcal{T}$ sees the highest bid that is not from themselves is $\beta_{\mathcal{B}}$, then the builder $\mathcal{T}$ infers that $v_{\mathcal{B}} = (b^1_{\mathcal{B}})^{-1}(\beta_{\mathcal{B}})$.

With these updated valuations (beliefs), the second round becomes a standard auction with valuations as common knowledge. Then it is easy to see that the builder $\mathcal{B}$ bidding $\min(v_{\mathcal{B}}, (b^1_{\mathcal{T}})^{-1}(\beta_{\mathcal{T}}))$ and the builder $\mathcal{T}$ bidding $\min(v_{\mathcal{T}}, (b^1_{\mathcal{B}})^{-1}(\beta_{\mathcal{B}}))$ is a Nash equilibrium. Using backward induction, we need to solve the Bayesian Nash equilibrium for the first round. In the next section, we formally describe the Bayesian subgame-perfect Nash equilibrium.

[2] We note that in the current MEV-Boost auction design, builders and the proposer can only query the highest bid, thus not exactly the same as our model. However, only revealing the highest bid would result in a *partial information update*, after which the auction becomes asymmetric. For asymmetric Bayesian auctions, the Bayesian Nash equilibrium has extremely complicated analytic forms, which are from differential equations [13]. Moreover, perhaps more importantly, for the sake of illustrating the effect of shill bidding, it is better for one to choose a reasonable setting that is simpler to understand, which can highlight the analysis around shill bidding better.

3.3 Bayesian subgame-perfect Nash equilibrium

We provide the characterization of the Bayesian subgame-perfect Nash equilibrium below.

▶ **Theorem 3.** *Assume $z \in [0, 1)$, $v_{\mathcal{B}} \sim U[0, \frac{1}{1-z}]$, and $v_{\mathcal{T}} \sim U[0, \frac{1}{1+z}]$. Then the following strategy profile is a Bayesian subgame-perfect Nash equilibrium:*

- *At the first round, $b^1_{\mathcal{B}} = b^*_{\mathcal{B}}(v_{\mathcal{B}})$ and $b^1_{\mathcal{T}} = b^*_{\mathcal{T}}(v_{\mathcal{T}})$, where $b^*_{\mathcal{B}}(\cdot)$ and $b^*_{\mathcal{T}}(\cdot)$ are the Bayesian Nash equilibrium strategy for the single round auction setting.*
- *At the second round (if it occurs), $b^2_{\mathcal{B}} = b^2_{\mathcal{T}} = \min(v_{\mathcal{B}}, v_{\mathcal{T}})$.*

Proof. The proof is by backward induction. Note that after the first round, each builder can see the highest bid from the other builder, which reveals their true valuation. Thus, in the second round, the bidding strategy under Nash equilibrium is just to bid $\min(v_{\mathcal{B}}, v_{\mathcal{T}})$. This is true regardless of the bidding strategies in the first round, as long as the builders use a strictly increasing bidding function. As a result, the builders should use the Bayesian Nash equilibrium strategy for the first round, which we can prove to be strictly increasing. ◀

We note that the Bayesian Nash equilibria for one-round auctions are known from the literature, and we include them below.

▶ **Theorem 4** (Vickrey [28], Maskin and Riley [14])**.** *When $z = 0$ and $v_{\mathcal{B}}, v_{\mathcal{T}} \sim [0, 1]$, we have*

$$b^*_{\mathcal{B}}(v_{\mathcal{B}}) = \frac{1}{2}v_{\mathcal{B}} \quad \text{and} \quad b^*_{\mathcal{T}}(v_{\mathcal{T}}) = \frac{1}{2}v_{\mathcal{T}}.$$

When $z \in (0, 1)$, $v_{\mathcal{B}} \sim [0, \frac{1}{1-z}]$, and $v_{\mathcal{T}} \sim [0, \frac{1}{1+z}]$, we have

$$b^*_{\mathcal{B}}(v_{\mathcal{B}}) = \frac{-1 + \sqrt{1 + 4v^2_{\mathcal{B}}z}}{4v_{\mathcal{B}}z} \quad \text{and} \quad b^*_{\mathcal{T}}(v_{\mathcal{T}}) = \frac{1 - \sqrt{1 - 4v^2_{\mathcal{T}}z}}{4v_{\mathcal{T}}z}.$$

It is important to note that this bidding function is strictly increasing, and for any $z \in [0, 1)$, we have $b^*_{\mathcal{B}}(\frac{1}{1-z}) = b^*_{\mathcal{T}}(\frac{1}{1+z}) = \frac{1}{2}$. Namely, both builders bid in the range $[0, \frac{1}{2}]$.

3.4 Effectiveness of shill bidding

Now we have understood the equilibrium bidding behavior of the builders. In this section, we aim to understand how the shill bidding will affect the bids.

The shill bidding strategy can be implemented as follows. In the first round, the proposer, who adopts the shill bidding strategy, may submit a shill bid. Note that this doesn't change the bidding behavior of any builder in the first round. However, after the first round, each builder collects all the bids that are not from themselves and regards them as the other builder's bids. In particular, this includes the shill bid. This makes them obtain misinformation about the other builder's valuation. Suppose that the shill bid in the first round is $\beta = \frac{1}{2}$. Note that it must be the highest among all the bids since both builders bid in the range $[0, \frac{1}{2}]$. Then, the builder $\mathcal{B}$ will refer that $v'_{\mathcal{T}} = (b^1_{\mathcal{T}})^{-1}(\beta) = \frac{1}{1+z}$. Similarly, the builder $\mathcal{T}$ will infer that $v'_{\mathcal{B}} = (b^1_{\mathcal{B}})^{-1}(\beta) = \frac{1}{1-z}$. As a result, the maximal bid is changed from $\min(v_{\mathcal{B}}, v_{\mathcal{T}})$ to $\max(\min(v_{\mathcal{B}}, \frac{1}{1+z}), \min(v_{\mathcal{T}}, \frac{1}{1-z}))$.

We first theoretically characterize the effectiveness of shill bidding for the symmetric case $v_{\mathcal{B}} \sim [0, 1]$ and $v_{\mathcal{T}} \sim [0, 1]$. Besides being interesting on its own, it can help demonstrate the effectiveness of shill bidding and illustrate the analysis methodology for the more general case later.

▶ **Theorem 5.** *When $v_{\mathcal{B}} \sim [0, 1]$ and $v_{\mathcal{T}} \sim [0, 1]$, we have:*

- *The proposer's expected revenue in the first round is* $1/3$*;*
- *The proposer's expected revenue in the second round without shill bidding is* $1/3$*;*
- *The proposer's expected revenue in the second round with shill bidding is* $2/3$*; and*
- *The expected maximal value of builders is* $2/3$*.*

Proof. For the first round, we know that the equilibrium bidding strategies are $b^1_{\mathcal{B}} = \frac{1}{2}v_{\mathcal{B}}$ and $b^1_{\mathcal{T}} = \frac{1}{2}v_{\mathcal{T}}$. Thus, we have

$$\mathbb{E}[\max(b^1_{\mathcal{B}}, b^1_{\mathcal{T}})] = \frac{1}{2}\int_0^1\int_0^1 \max(v_{\mathcal{B}}, v_{\mathcal{T}})\, dv^1_{\mathcal{T}}\, dv^1_{\mathcal{B}} = \frac{1}{3}.$$

For the second round, if there is no shill bidding, we know that the equilibrium bidding strategies are $b^2_{\mathcal{B}} = b^2_{\mathcal{T}} = \min(v_{\mathcal{B}}, v_{\mathcal{T}})$. Thus, we have

$$\mathbb{E}[\min(v_{\mathcal{B}}, v_{\mathcal{T}})] = \int_0^1\int_0^1 \min(v_{\mathcal{B}}, v_{\mathcal{T}})\, dv^1_{\mathcal{T}}\, dv^1_{\mathcal{B}} = \frac{1}{3}.$$

For the second round, if there *is* shill bidding, we know that the bids of both builders are $b^2_{\mathcal{B}} = \min(v_{\mathcal{B}}, 1)$ and $b^2_{\mathcal{T}} = \min(v_{\mathcal{T}}, 1)$. Thus, we have

$$\mathbb{E}[\max(\min(v_{\mathcal{B}}, 1), \min(v_{\mathcal{T}}, 1))] = \int_0^1\int_0^1 \max(v_{\mathcal{B}}, v_{\mathcal{T}})\, dv^1_{\mathcal{T}}\, dv^1_{\mathcal{B}} = \frac{2}{3}.$$

The expected maximal value of builders is

$$\mathbb{E}[\max(v_{\mathcal{B}}, v_{\mathcal{T}})] = \int_0^1\int_0^1 \max(v_{\mathcal{B}}, v_{\mathcal{T}})\, dv^1_{\mathcal{T}}\, dv^1_{\mathcal{B}} = \frac{2}{3}.$$

This finishes the proof. ◀

We have successfully analyzed the important quantities to study the effectiveness of shill bidding for the classic symmetric case. For the more general case $z \in (0, 1)$, however, the analytic solution needs to solve a very complicated integral, which may not be closed-form. Thus, instead, we provide the integral formula of all the quantities of interest (the proposer's expected revenue in the first round, the proposer's expected revenue in the second round without shill bidding, the proposer's expected revenue in the second round with shill bidding, and the expected maximal value of builders), and use numerical experiments to approximate the results with demonstrations in Figure 1 and Figure 2. As we are going to discuss in the next section, these approximations and demonstrations are clear enough for us to get important insights about the effectiveness of shill bidding.

Integral forms of the quantities of interest. We summarize the integral forms below.

- Proposer's expected revenue in the first round:

$$\int_0^{\frac{1}{1-z}}\int_0^{\frac{1}{1+z}} \max\left(\frac{1-\sqrt{1-4v_{\mathcal{T}}^2 z}}{4v_{\mathcal{T}}z}, \frac{-1+\sqrt{1+4v_{\mathcal{B}}^2 z}}{4v_{\mathcal{B}}z}\right) dv^1_{\mathcal{T}}\, dv^1_{\mathcal{B}}.$$

- Proposer's expected revenue in the second round without shill bidding:

$$\int_0^{\frac{1}{1-z}}\int_0^{\frac{1}{1+z}} \min\left(v_{\mathcal{T}}, v_{\mathcal{B}}\right)\, dv^1_{\mathcal{T}}\, dv^1_{\mathcal{B}}.$$

- Proposer's expected revenue in the second round with shill bidding:

$$\int_0^{\frac{1}{1-z}}\int_0^{\frac{1}{1+z}} \max\left(\min\left(v_{\mathcal{T}}, \frac{1}{1-z}\right), \min\left(v_{\mathcal{B}}, \frac{1}{1+z}\right)\right) dv^1_{\mathcal{T}}\, dv^1_{\mathcal{B}}.$$

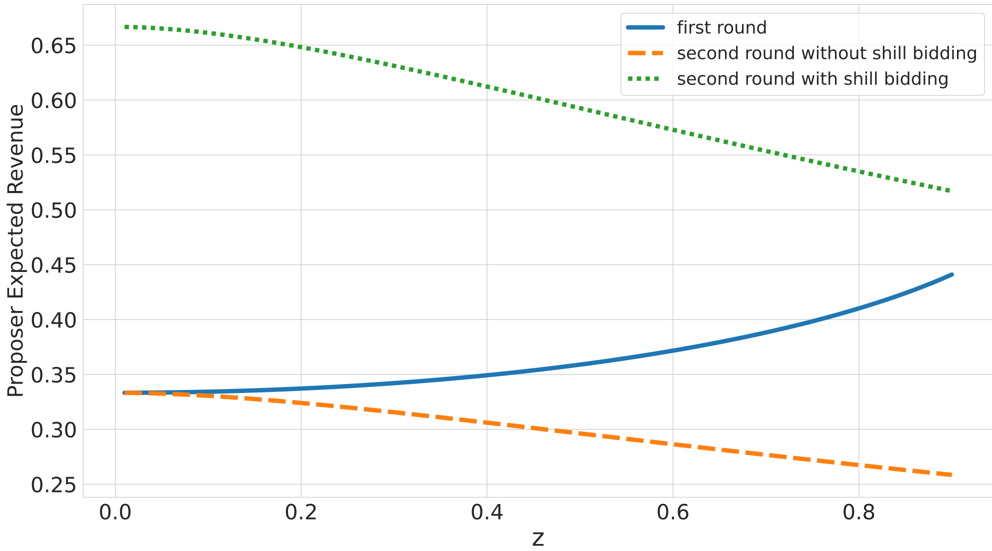

Figure 1 Proposer's expected revenue in the first round, second round (with/without shill bidding), with respect to z.

- Expected maximal value of builders:

$$\int_0^{\frac{1}{1-z}} \int_0^{\frac{1}{1+z}} \max\left(v_{\mathcal{T}}, v_{\mathcal{B}}\right)\, dv_{\mathcal{T}}^1\, dv_{\mathcal{B}}^1.$$

3.5 Interpretation

In this section, we interpret the numerical results obtained in the previous section. We divide the discussion into three parts.

Effectiveness of shill bidding compared to non-shill bidding. As shown in Figure 1, the proposer's expected revenue with shill bidding in the second round (green line) is twice as high as the revenue without shill bidding (orange line), as reflected by the numerical values (and also almost clear from the figure). Thus, it would not be a surprise to recall our Theorem 5, which also proved such a twice-improvement on the proposer's expected revenue. This is somewhat surprising since it holds even when z grows, which means the competitiveness of the auction decreases.

Effectiveness of shill bidding compared to the maximal value of the builders. Now, let's focus on Figure 2, which adds a line for the expected maximal value of builders based on Figure 1. As shown in Figure 2, the proposer's expected revenue with shill bidding in the second round (green line) is almost the same as the maximal value of the builders (red line) when z is small. However, when z is large, there is a significant gap between these two quantities. This means that when the two builders have basically the same capacity to extract MEV, shill bidding is super effective, in the sense that it can help the proposer obtain almost all the revenue (whereas without shill bidding, the proposer is only able to obtain half of that). However, when there is a dominant builder, shill bidding is not as effective as expected. Beaver recently announced its full migration to BuilderNet (a decentralized block-building platform for Ethereum), which is currently in progress. Hypothetically, for

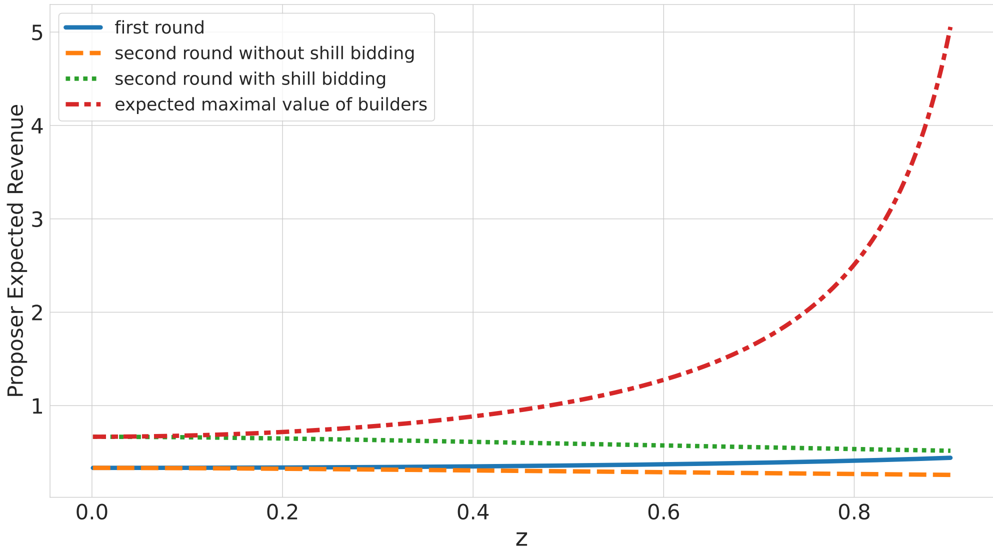

Figure 2 Proposer's expected revenue in the first round, second round (with/without shill bidding), and the expected maximal value of builders, with respect to z. This is to demonstrate the huge gap between the expected maximal value of builders and the other three, especially when z is large.

example, if Titan also joins BuildNet in the future, then shill bidding would not be able to reduce the huge gap in the proposer's loss anymore.

The intuition behind the phenomenon above is as follows: Suppose that there is a dominant builder, whose valuation is usually, say, fifty times that of other builders' valuations. Then, if seeing an abnormally high bid, the dominant builder would be able to realize something strange had happened, and they could choose to ignore such a bid. Thus, shill bidding cannot be effective enough. Oppositely, if there are multiple builders whose abilities are roughly the same, the key observation is that they cannot really distinguish whether a relatively high bid is from the other builders or it is a shill bid. Thus, they may choose to respond by bidding high, which makes the shill bidding very effective.

The effect of shill bidding on timing games. As shown in Figure 2, the proposer's expected revenue in the second round, absent shill bidding, is lower than that in the first round. Therefore, under our model, the proposer would have no incentive to engage in timing games, namely, delaying the termination of the MEV-Boost auction.[3] However, with shill bidding, the proposer's expected revenue in the second round exceeds that of the first round. This introduces unexpected incentives to delay the termination of the MEV-Boost auction, making timing games more appealing to the proposer. We regard this as one of the most concerning consequences of shill bidding.

[3] It is important to note that our setting does not incorporate the feature wherein value increases over time – a key characteristic of MEV-Boost auctions and a primary driver of timing games. Our focus, instead, is to illustrate that shill bidding can raise the expected second-round revenue, thereby incentivizing the proposer to delay the auction termination even in the absence of time-dependent valuations.

Table 1 Overview of collected datasets.

Dataset	Content	# of entries	Source
On-chain	blocks	6,210,556	Reth node [20]
	transactions	966,644,462	
Auction	blocks produced by MEV-Boost	5,510,175	relay API
	partial bids	13,994,696,728	relay API
	full bids from ultra sound	326,574,676	ultra sound
Misc.	builder public keys	295	building on [34]

4 Empirical Study

In this section, we study shill bidding using empirical data from MEV-Boost auctions on Ethereum. Specifically, we aim to answer the following research questions:

1. Is there sufficient incentive for proposers to engage in the shill bidding strategy in practice?
2. Are proposers actually adopting the shill bidding strategy in practice?

4.1 Data sets

We curated the following datasets to support the empirical study, as summarized in Table 1.

On-chain data. We run a local Ethereum node (Reth [20]) to collect 6,210,556 blocks and 966,644,462 transactions on Ethereum from September 15, 2022 to January 31, 2025.

Data from MEV-Boost auctions. We connected to all active relays and collected two types of data from their public APIs [7] between September 15, 2022, and January 31, 2025:

- Winning bids, which include 5,510,175 finalized blocks sent to proposers. We use these to identify blocks produced via MEV-Boost auctions.
- Partial bids (non-winning bids without block bodies) via the relay API. The dataset contains 13,994,696,728 partial bids, each consisting of a block hash, bid value, and builder public key. We use these to analyze builders' behavior.

To further bolster our analysis, we also use the full bids dataset collected by prior work [34], which recorded 301,479 historical MEV-Boost auctions between April and December 2023. The dataset contains 326,574,676 full bids, each consisting of a block (rather than the block hash), bid value, and builder public key. The representativeness of this dataset has been validated by [34].

Miscellaneous data. Builders are often identified by name (e.g., Flashbots) but may use multiple public keys to sign bids. Therefore, to attribute bids in an MEV-Boost auction to builders, a mapping from public keys to builder identities is required. We construct such a mapping, which includes 295 builder public keys, based on prior work [34].

4.2 Is shill bidding worth adopting?

In this section, we address the first research question: whether it is worth adopting shill bidding for proposers. To investigate this, we estimate the potential additional revenue proposers could gain through shill bidding. Specifically, we simulate a scenario where the proposer adopts shill bidding, causing the original winner to still win the auction, but to bid their true valuation.

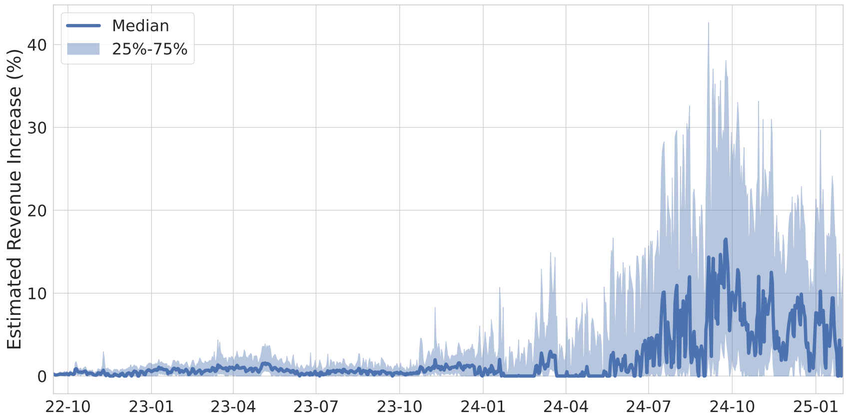

Figure 3 Estimated percentage increase in proposer revenue per MEV-Boost auction under the shill bidding strategy.

We follow the same approach as the previous works [18, 34] to compute the true value of the winner's bid. Concretely, we calculate the sum of all transaction fees paid by transactions within a block and the ETH transferred to the coinbase.

Assuming shill bidding compels the original winner to bid their full true value, we calculate the difference between this true value and the original winning bid. This difference reflects the additional revenue available to the proposer in each MEV-Boost auction slot. We investigate the additional revenue for proposers from two distinct dimensions:

- Relative Value: This is quantified as the percentage increase in revenue that each proposer could gain in individual MEV-Boost auctions, relative to their prior earnings.
- Absolute Value: This represents the aggregate additional revenue accumulated by proposers across all MEV-Boost auctions.

Figure 3 provides a continuous visualization of the relative additional revenue achievable by proposers from September 2022 to January 2025. The solid blue line in the figure represents the median of this percentage revenue increase, calculated daily across all MEV-Boost auctions. This is complemented by its interquartile range (25th to 75th percentile), which is depicted by the shaded light blue area.

We can make two observations from the figure: First, in 2022 and 2023, the percentage revenue increase remained moderate, with the daily 75th percentile generally not exceeding 5%. Second, the percentage revenue increase becomes significantly more pronounced from 2024 onwards, where the daily median frequently exceeds 15% and the daily 75th percentile can surpass 40%. These observations collectively indicate that employing a shill bidding strategy to compel winners to increase their bid can lead to a highly substantial growth in proposers' revenue, particularly evident from 2024 onwards. This highlights a significant increase in the potential financial incentives for proposers leveraging this strategy.

As shown in Figure 4, throughout 2022 and 2023, the potential additional revenue for proposers was not markedly substantial, with monthly values consistently remaining below 3,000 ETH. In stark contrast, from January 2024 onwards, the monthly potential additional revenue demonstrated a significant escalation, consistently exceeding 3,000 ETH and peaking at over 7,000 ETH in August 2024. This notable shift aligns consistently with trends observed in Figure 3, particularly reinforcing the finding that from 2024 onwards (and projecting to the time of the writing), adopting a shill bidding strategy can generate substantial additional monthly revenue for proposers.

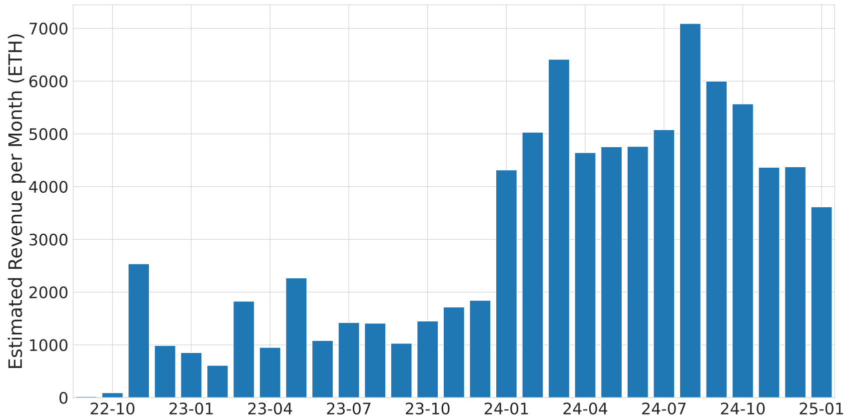

Figure 4 Estimated additional revenue per month for proposers in MEV-Boost auctions when using the shill bidding strategy.

4.3 Are proposers adopting shill bidding?

In this section, we aim to answer the second research question: Have proposers already adopted the shill bidding in practice? To this end, we propose two heuristic methods to identify potential evidence of shill bidding in past MEV-Boost auctions. Each heuristic is designed based on behavioral signatures that may arise if proposers were engaging in such strategic behavior.

- Heuristic 1: If proposers want to adopt a shill bidding strategy, they can submit bids significantly higher than their true values. That is, the presence of a bid significantly higher than the block's true value may indicate potential shill bidding activity.
- Heuristic 2: A proposer can request the highest bid from the relay multiple times and eventually sign a high-enough real bid. That is, the existence of an eligible bid higher than the winning bid suggests that the proposer is strategically ignoring shill bids from their affiliated builders.

Results for Heuristic 1. We applied the first heuristic to our full bids dataset to examine the presence of significantly inflated bids. More specifically, for each MEV-Boost auction in the dataset, we check whether any bid exceeded the corresponding block's true value by a specific threshold. We considered three thresholds: a bid value at least twice the true value, at least 1.5 times the true value, and at least 4/3 of the true value. To avoid false positives stemming from builder subsidies, a common strategy where a builder bids slightly above the true value to gain market share [18, 34], we excluded cases where the excess amount is small (specifically, less than or equal to 0.02 ETH, as estimated from an online dashboard [22]).

Across the 301,479 MEV-Boost auctions in our full bids dataset, we found no bids meeting the criteria of Heuristic 1. This suggests that, during the period covered by the dataset, there is no clear evidence that proposers employed shill bidding by submitting bids significantly above the true value.

Results for Heuristic 2. Due to the limited temporal coverage of the full bids dataset (restricted to 2023), we extend our analysis by applying the second heuristic to our partial bids dataset, which is significantly larger and spans MEV-Boost auctions from 2022 to 2025. For each auction in the dataset, we check whether any eligible bid offered a higher value than the winning bid but failed to be selected. We focus on optimistic submissions, namely, bids that are eligible for immediate return to proposers upon receipt by the relay, thereby incurring no validation delay [16].

In our experiments, we identify 666,808 out of 5,510,175 auctions where such a higher bid exists. However, such higher bids were submitted *before* the winning bid, suggesting that this may be caused by an operational error on the relay side. In 64.2% of these cases, the earlier, higher bid originates from the same relay as the winning bid and is ultimately attributed to the eventual winner, although under a different builder public key. In the remaining cases, the higher bid originates from a (usually well-known) builder distinct from the eventual winner.

Given that a builder has no incentive to engage in shill bidding against themselves, and that relays are expected to forward the highest eligible bid (which does not occur here), we hypothesize that these cases are most likely due to a bug or misconfiguration in the relay operations. This finding has been reported to the relevant relays for further investigation. Due to this potential bug, we cannot conclusively determine whether proposers have actively adopted shill bidding based on Heuristic 2. Note that the potential issue in the relay does not affect our results in Section 4.2, which only rely on the on-chain data.

Limitations. While our empirical analysis did not uncover direct evidence of shill bidding, the following considerations help contextualize this result and explain the challenges involved:

First, as shown in Section 4.2, adopting a shill bidding strategy could offer proposers substantial revenue gains. The fact that these potential gains remain large suggests that such strategies may not yet be widely adopted; were they already widespread, we would expect the marginal benefits from adopting them to diminish due to increased competition.

Second, the accurate analysis of behaviors indicative of shill bidding depends critically on trustworthy relay data. Our findings, however, have uncovered potential misconfigurations within the relay. This concern makes it challenging to ascertain definitively whether observed patterns truly constitute shill bidding by proposers or are artifacts of these relay issues.

Finally, it is plausible that real-world shill bidding strategies, if employed, are more complicated or adaptive than those encompassed by our current heuristics.

5 Conclusion and Discussion

We proposed and investigated a novel shill bidding strategy in MEV-Boost auctions, where the proposer inflates bids to manipulate the selling price. Our theoretical analysis and numerical results offer valuable insights into the effectiveness of this strategy from multiple perspectives. Our empirical results show that this strategy could significantly increase proposer revenue, particularly from 2024 onward. This highlights a growing incentive for such behavior, while also suggesting that it is not currently being adopted, although we did not find conclusive evidence.

Although shill bidding is typically regarded as fraudulent in traditional auctions, in the context of the MEV-Boost auction, particularly given the current high level of vertical integration, shill bidding proposed in this paper may be interpreted as a defensive strategy employed by the proposer. Regardless, the potential advantages and drawbacks of such behavior deserve more attention from the community and further discussion.

Several interesting directions emerge when considering how various stakeholders, such as builders, relay, and protocol designers, might respond to shill bidding. We outline three promising areas for future exploration:

- *Coalition among builders.* To counter shill bidding, top builders can form a coalition to constantly synchronize their true valuations during the MEV-Boost auction. This requires careful mechanism design to ensure credible and secure information sharing. Without proper incentive alignment, coalitions risk breaking down: for instance, consider builders $\mathcal{B}$ and $\mathcal{T}$ forming a coalition; if one builder privately benefits from strategically misreporting values or timing their disclosures to gain individual advantage, the coalition could quickly unravel.

- *Commission fee by the relay.* Inspired by shill bidding literature in traditional auctions (e.g., [31]), one promising mechanism to mitigate shill bidding is the introduction of a carefully designed commission fee charged by the relay to the proposer. By imposing a cost on the final bid, such a fee may render shill bidding unprofitable for proposers. A relay implementing this mechanism could attract builders who prefer to participate in less manipulable auctions. Simultaneously, such fees could serve as a funding resource, helping the relay cover its operational and institutional expenses.
- *Revisiting MEV-Boost.* If the top builders in the market are relatively evenly matched, shill bidding becomes ineffective as builders naturally compete aggressively, driving bids to their truthful values. This insight raises the question of whether existing PBS implementation through MEV-Boost auctions can be modified or improved to inherently neutralize shill bidding incentives, e.g., by leveling the playing field among builders.

References

1. Maryam Bahrani, Pranav Garimidi, and Tim Roughgarden. Transaction Fee Mechanism Design in a Post-MEV World. In Rainer Böhme and Lucianna Kiffer, editors, *6th Conference on Advances in Financial Technologies (AFT '24), September 23-25, 2024, Vienna, Austria*, volume 316 of *LIPIcs*, pages 29:1–29:24. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2024. URL: `https://doi.org/10.4230/LIPIcs.AFT.2024.29`, `doi:10.4230/LIPICS.AFT.2024.29`.
2. Indranil Chakraborty and Georgia Kosmopoulou. Auctions with shill bidding. *Economic Theory*, 24:271–287, 2004.
3. Kong-Pin Chen, Ting-Peng Liang, Ted Chang, Yi-chun Liu, Shou-Yung Yin, and Ya-Ting Yu. How serious is shill bidding in online auctions? evidence from ebay motors. *Evidence from eBay Motors (September 26, 2020)*, 2020.
4. Hao Chung, Tim Roughgarden, and Elaine Shi. Collusion-resilience in transaction fee mechanism design. In *Proceedings of the 25th ACM Conference on Economics and Computation*, pages 1045–1073, 2024.
5. Philip Daian, Steven Goldfeder, Tyler Kell, Yunqi Li, Xueyuan Zhao, Iddo Bentov, Lorenz Breidenbach, and Ari Juels. Flash Boys 2.0: Frontrunning in Decentralized Exchanges, Miner Extractable Value, and Consensus Instability. In *2020 IEEE symposium on security and privacy (SP'20)*, pages 910–927. IEEE, 2020.
6. Ethereum. Proposer-Builder Separation. `https://ethereum.org/en/roadmap/pbs/`, 2025. Accessed: May 27, 2025.
7. Flashbots. Flashbots Relay API Specification. `https://flashbots.github.io/relay-specs/`, 2025. Accessed: May 27, 2025.
8. Yotam Gafni and Aviv Yaish. Barriers to collusion-resistant transaction fee mechanisms. In *Proceedings of the 25th ACM Conference on Economics and Computation*, pages 1074–1096, 2024.
9. Aadityan Ganesh, Clayton Thomas, and S. Matthew Weinberg. Revisiting the Primitives of Transaction Fee Mechanism Design. In *EC*, page 703. ACM, 2024.
10. Tivas Gupta, Mallesh M. Pai, and Max Resnick. The Centralizing Effects of Private Order Flow on Proposer-Builder Separation. In *5th Conference on Advances in Financial Technologies (AFT 2023)*, volume 282 of *LIPIcs*, pages 20:1–20:15. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2023.

11 Lioba Heimbach, Lucianna Kiffer, Christof Ferreira Torres, and Roger Wattenhofer. Ethereum's Proposer-Builder Separation: Promises and Realities. In *Proceedings of the 2023 ACM on Internet Measurement Conference*, pages 406–420, 2023.

12 Lioba Heimbach, Vabuk Pahari, and Eric Schertenleib. Non-Atomic Arbitrage in Decentralized Finance. In *2024 IEEE Symposium on Security and Privacy (SP)*, pages 3866–3884. IEEE, 2024.

13 Todd R Kaplan and Shmuel Zamir. Asymmetric first-price auctions with uniform distributions: analytic solutions to the general case. *Economic Theory*, 50(2):269–302, 2012.

14 Eric Maskin and John Riley. Asymmetric auctions. *The review of economic studies*, 67(3):413–438, 2000.

15 Mike Neuder, Thomas Thiery, and Chris Hager. Bid cancellations considered harmful. `https://ethresear.ch/t/bid-cancellations-considered-harmful/15500`, 2023. Accessed: May 27, 2025.

16 Michael Neuder. Optimistic relay proposal. `https://github.com/michaelneuder/optimistic-relay-documentation/blob/main/proposal.md`, 2022. Accessed: May 28, 2025.

17 Alexey Nikitkov and Darlene Bay. Shill bidding: Empirical evidence of its effectiveness and likelihood of detection in online auction systems. *International Journal of Accounting Information Systems*, 16:42–54, 2015.

18 Burak Öz, Danning Sui, Thomas Thiery, and Florian Matthes. Who wins ethereum block building auctions and why? In *6th Conference on Advances in Financial Technologies (AFT)*, 2024.

19 Mallesh Pai and Max Resnick. Structural Advantages for Integrated Builders in MEV-Boost. In *International Conference on Financial Cryptography and Data Security*, pages 128–132. Springer, 2024.

20 Paradigm. Reth: A pure Rust Ethereum execution layer. `https://github.com/paradigmxyz/reth`, 2025. Accessed: May 27, 2025.

21 Kaihua Qin, Liyi Zhou, Pablo Gamito, Philipp Jovanovic, and Arthur Gervais. An Empirical Study of DeFi Liquidations: Incentives, Risks, and Instabilities. In *Proceedings of the 21st ACM internet measurement conference*, pages 336–350, 2021.

22 RelayScan.io. RelayScan: The World's First Real-Time Relay Network Scan. `https://www.relayscan.io/`, 2025. Accessed: 2025-05-28.

23 Tim Roughgarden. Transaction Fee Mechanism Design. In Péter Biró, Shuchi Chawla, and Federico Echenique, editors, *Proceedings of the 22nd ACM Conference on Economics and Computation (EC '21)*, page 792. ACM, 2021. `doi:10.1145/3465456.3467591`.

24 Caspar Schwarz-Schilling, Fahad Saleh, Thomas Thiery, Jennifer Pan, Nihar Shah, and Barnabé Monnot. Time Is Money: Strategic Timing Games in Proof-Of-Stake Protocols. In *5th Conference on Advances in Financial Technologies (AFT)*, volume 282 of *LIPIcs*, pages 30:1–30:17. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2023.

25 Thomas Thiery. Empirical analysis of Builders' Behavioral Profiles (BBPs). `https://ethresear.ch/t/empirical-analysis-of-builders-behavioral-profiles-bbps/16327`, 2023. Accessed: May 27, 2025.

26 Toni Wahrstätter. Ethereum Timing Dashboard. `https://timing.pics/`, 2025. Accessed: May 25, 2025.

27 Toni Wahrstätter. MEV-Boost Dashboard. `https://mevboost.pics/`, 2025. Accessed: May 27, 2025.

28 William Vickrey. Counterspeculation, auctions, and competitive sealed tenders. *The Journal of finance*, 16(1):8–37, 1961.

29 Anton Wahrstätter, Jens Ernstberger, Aviv Yaish, Liyi Zhou, Kaihua Qin, Taro Tsuchiya, Sebastian Steinhorst, Davor Svetinovic, Nicolas Christin, Mikolaj Barczentewicz, et al. Blockchain Censorship. In *Proceedings of the ACM Web Conference 2024*, pages 1632–1643, 2024.

30 Shuzheng Wang, Yue Huang, Wenqin Zhang, Yuming Huang, Xuechao Wang, and Jing Tang. Private order flows and builder bidding dynamics: The road to monopoly in ethereum's block building market. In *Proceedings of the ACM on Web Conference 2025*, pages 2144–2157, 2025.

31 Wenli Wang, Zoltán Hidvégi, and Andrew B Whinston. Shill Bidding in English Auctions. *Emory University and the University of Texas at Austin, mimeo*, 2001.

32 Ye Wang, Yan Chen, Haotian Wu, Liyi Zhou, Shuiguang Deng, and Roger Wattenhofer. Cyclic Arbitrage in Decentralized Exchanges. In *Companion Proceedings of the Web Conference 2022*, pages 12–19. ACM, 2022.

33 Fei Wu, Thomas Thiery, Stefanos Leonardos, and Carmine Ventre. Strategic Bidding Wars in On-chain Auctions. In *2024 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, pages 503–511. IEEE, 2024.

34 Sen Yang, Kartik Nayak, and Fan Zhang. Decentralization of Ethereum's Builder Market. In *2025 IEEE Symposium on Security and Privacy (SP)*, pages 1512–1530, Los Alamitos, CA, USA, 2025. IEEE, IEEE Computer Society.

35 Sen Yang, Fan Zhang, Ken Huang, Xi Chen, Youwei Yang, and Feng Zhu. Sok: Mev countermeasures: Theory and practice. *arXiv preprint arXiv:2212.05111*, 2022.

36 Liyi Zhou, Kaihua Qin, Christof Ferreira Torres, Duc V Le, and Arthur Gervais. High-Frequency Trading on Decentralized On-Chain Exchanges. In *2021 IEEE Symposium on Security and Privacy (SP'21)*, pages 428–445. IEEE, 2021.